SCHOOL OF NATURAL AND APPLIED SCIENCES

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PHASE 3

Graphical User Interface and Database Design

Field	Details
Name of Document	Phase 3: Graphical User Interface and Database Design
Project Title	AgroAdvisor: an AI-assisted farm advisory platform for smallholder farmers
Name of the System	AgroAdvisor
Student Name	[Student name]
Student Number	[Student number]
Submission Date	[Confirm submission date with the supervisor]

Yellow-highlighted text marks content that still needs a personal detail or real evidence from the student.

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1. Graphical User Interface Design

This section shows every screen of the working AgroAdvisor system. The screenshots were taken from the running application on a phone-sized layout, because most farmers use phones. Long pages are cut into side-by-side panels that read left to right, then top to bottom. The logged-in screens show a sample farmer account with sample data.

Common design features on all screens:

- Earthy theme: dark and light greens with warm sand and soil tones, defined once as theme tokens.
- A header with the logo, a language selector (five languages) and a simple navigation bar: Dashboard, My farm, Learn, Farm memory, Community map, Log out.
- Clear loading, empty and error messages and labelled form fields.

1.1 Home page

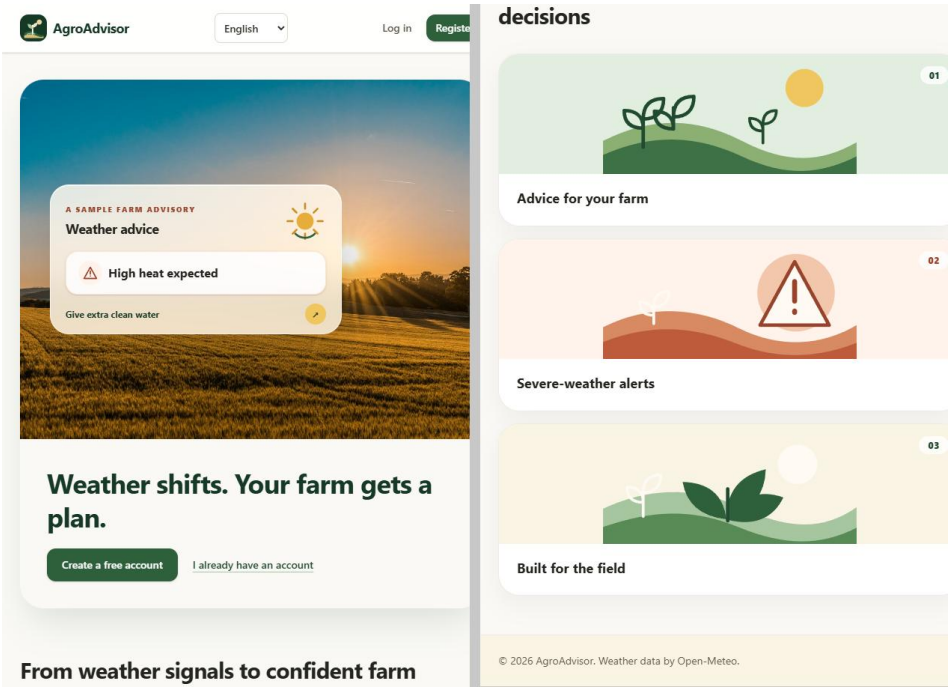


Figure: Home page (public)

- Purpose: introduces AgroAdvisor to a visitor before sign-up.
- Shows: a hero photo with a sample farm advisory card, the promise 'Weather shifts. Your farm gets a plan.', and three feature cards (advice for your farm, severe-weather alerts, built for the field).
- Actions: Create a free account; I already have an account; change language.

1.2 Register page

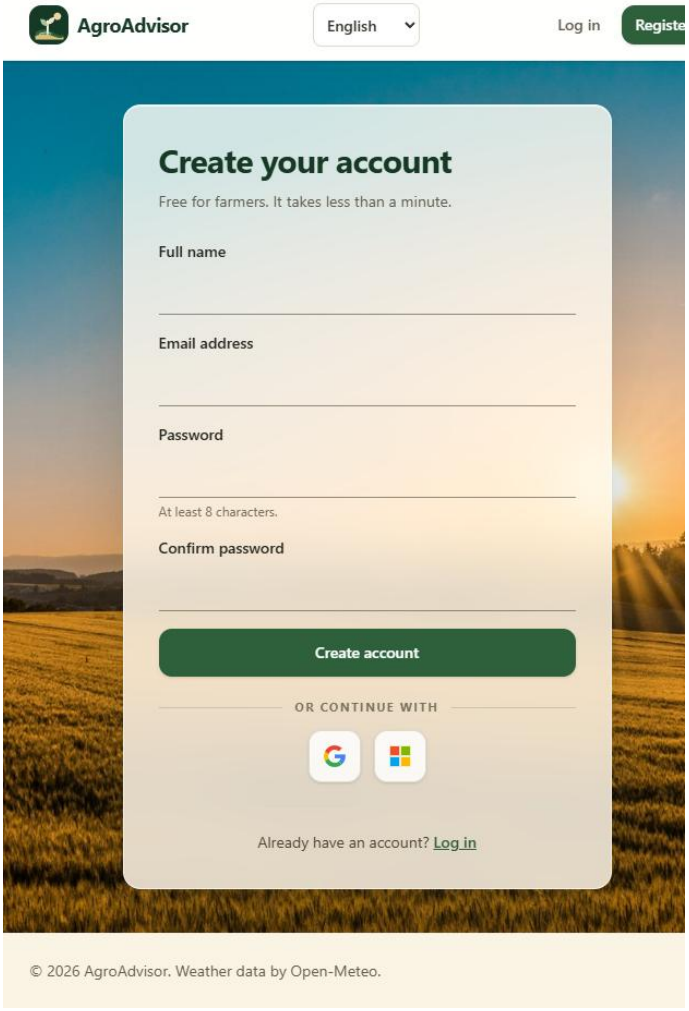


Figure: Register page

- Purpose: creates a farmer account.
- Fields: full name, email address, password (at least 8 characters) and confirm password.
- Actions: Create account; continue with Google or Microsoft; link to log in. Errors appear under the form in plain language.

1.3 Login page

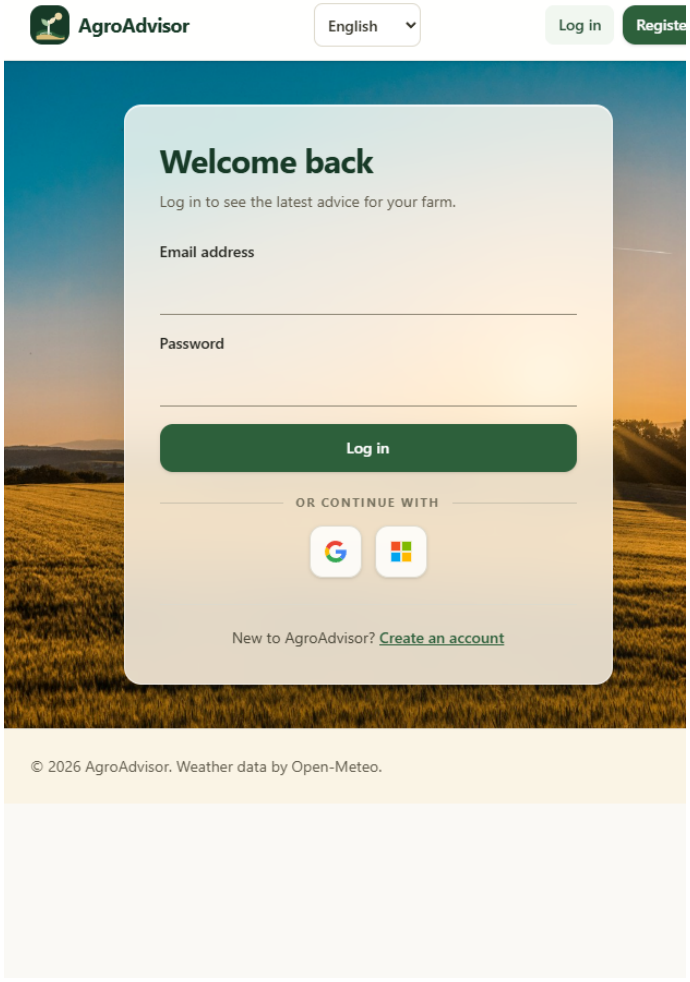


Figure: Login page

- Purpose: lets a returning farmer sign in.
- Fields: email address and password.
- Actions: Log in; continue with Google or Microsoft; link to create an account.

1.4 Dashboard

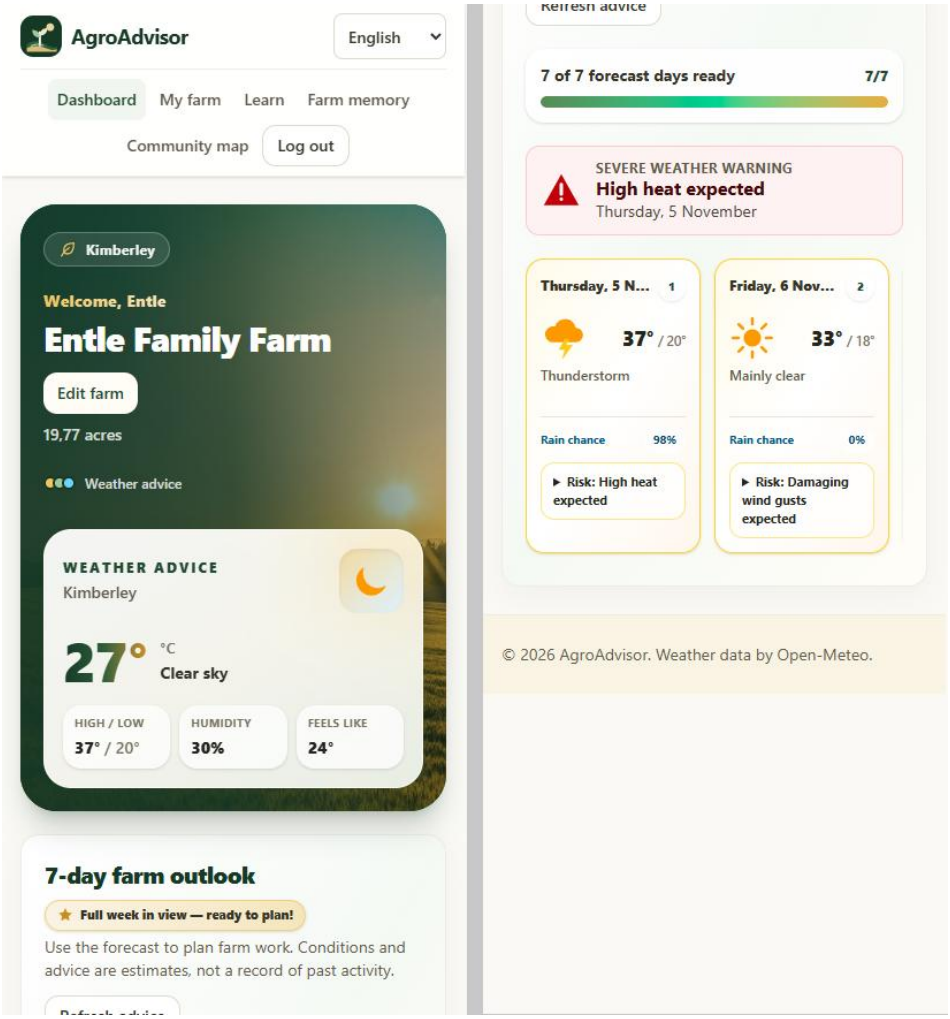


Figure: Dashboard: welcome banner, weather advice and 7-day farm outlook

- Purpose: the farmer's daily view. It turns the forecast into advice for this farm.
- Top banner: welcome with the farmer's name, farm name, size, nearest town and an Edit farm button. A weather advice card next to it shows the current temperature, conditions, high and low, humidity and feels-like.
- 7-day farm outlook: a progress bar of forecast days ready, a severe-weather warning strip (for example 'High heat expected' with the date), and one card per day with conditions, temperature range, rain chance and a risk note.
- Actions: Edit farm; Refresh advice; open a warning to read what to do and play it as audio.

1.5 My farm

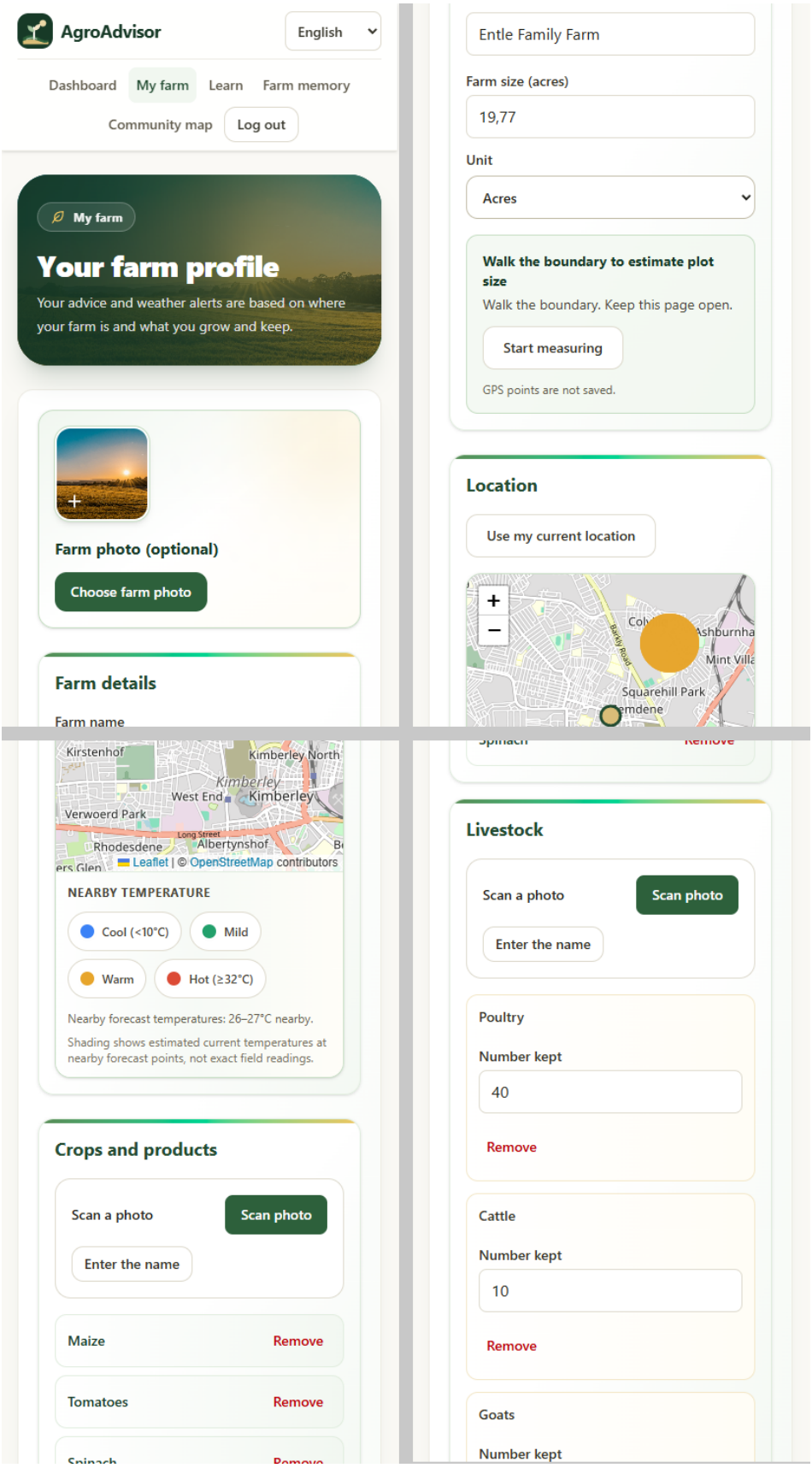


Figure: My farm: farm profile form

- Purpose: stores the information that advice is based on.

- Sections: farm photo (optional); farm details (name, size with acres or hectares); walk-the-boundary plot measurement; location map with a button to use the current location and a nearby-temperature layer (cool, mild, warm, hot); crops and products; livestock with a number kept for each animal.
- Crops and animals can be added by scanning a photo or by typing the name, and removed with one tap.
- Action: Save changes.

1.6 Learn

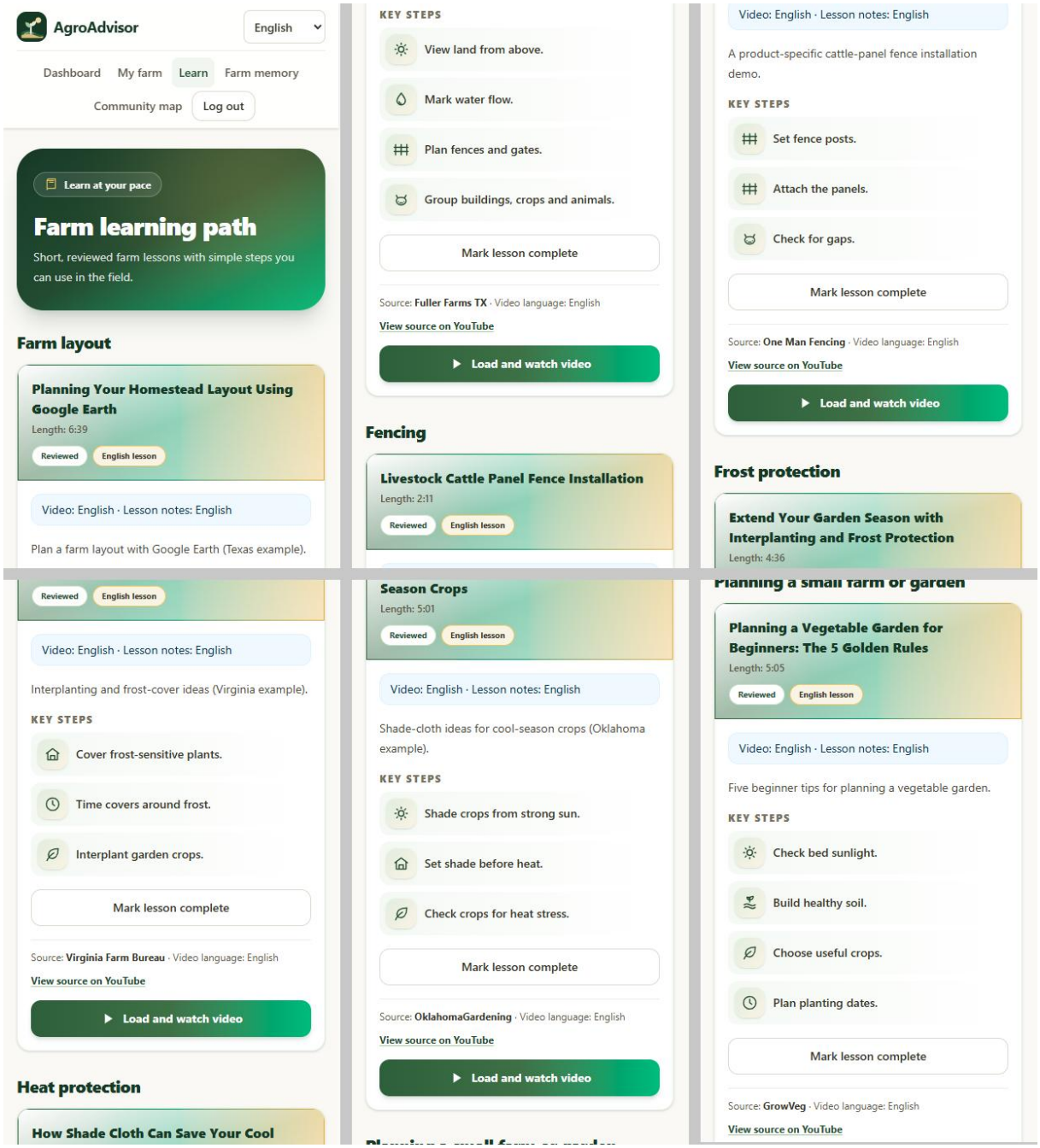


Figure: Learn: farm learning path

- Purpose: short reviewed video lessons that farmers can use in the field.

- Shows: lessons grouped by topic (farm layout, fencing, frost protection, heat protection, planning a small farm). Each card has the title, length, a Reviewed badge, a language note, a short summary, key steps with icons and the video source.
- Actions: Load and watch video (the video loads only when pressed, which saves mobile data); Mark lesson complete; open the source on YouTube.

1.7 Farm memory

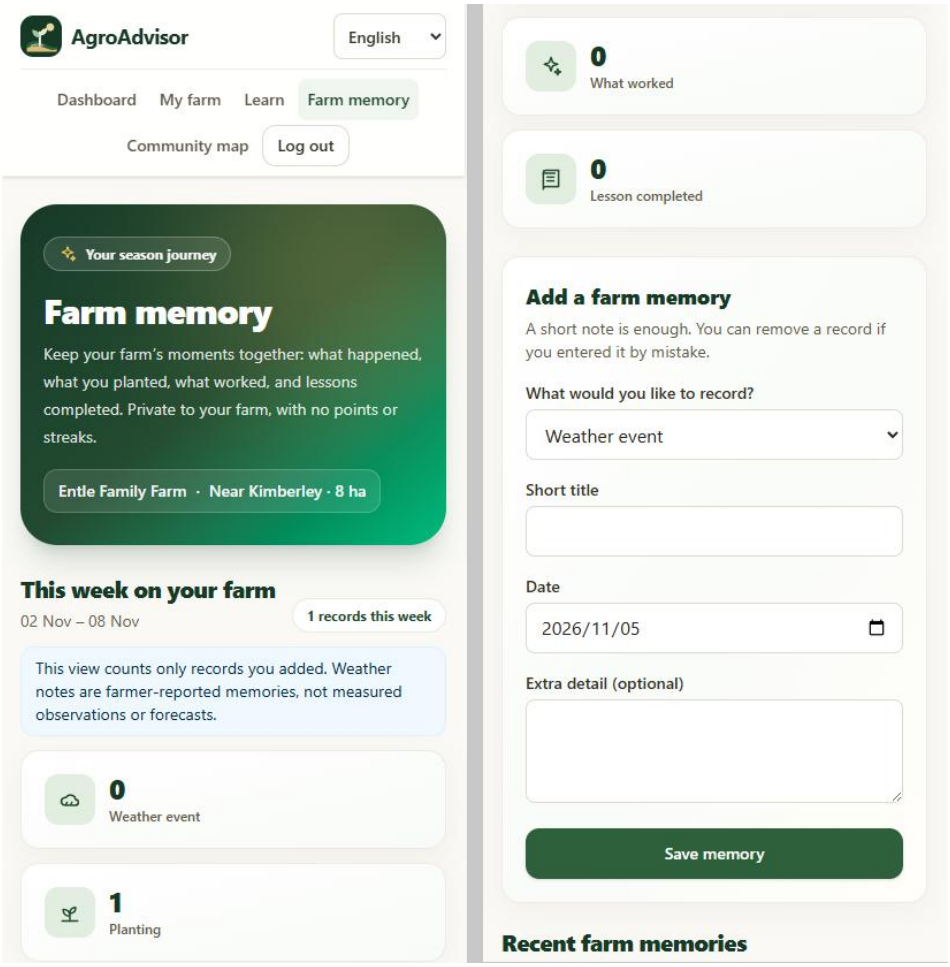


Figure: Farm memory: weekly summary and add a memory

- Purpose: a private record of what happened on the farm.
- Shows: this week's counts for weather events, plantings, what worked and lessons completed, then a form and a list of recent memories.
- Add a memory: choose the type, write a short title, pick a date and add an optional detail, then Save memory. A record can be removed if it was entered by mistake.

1.8 Community map

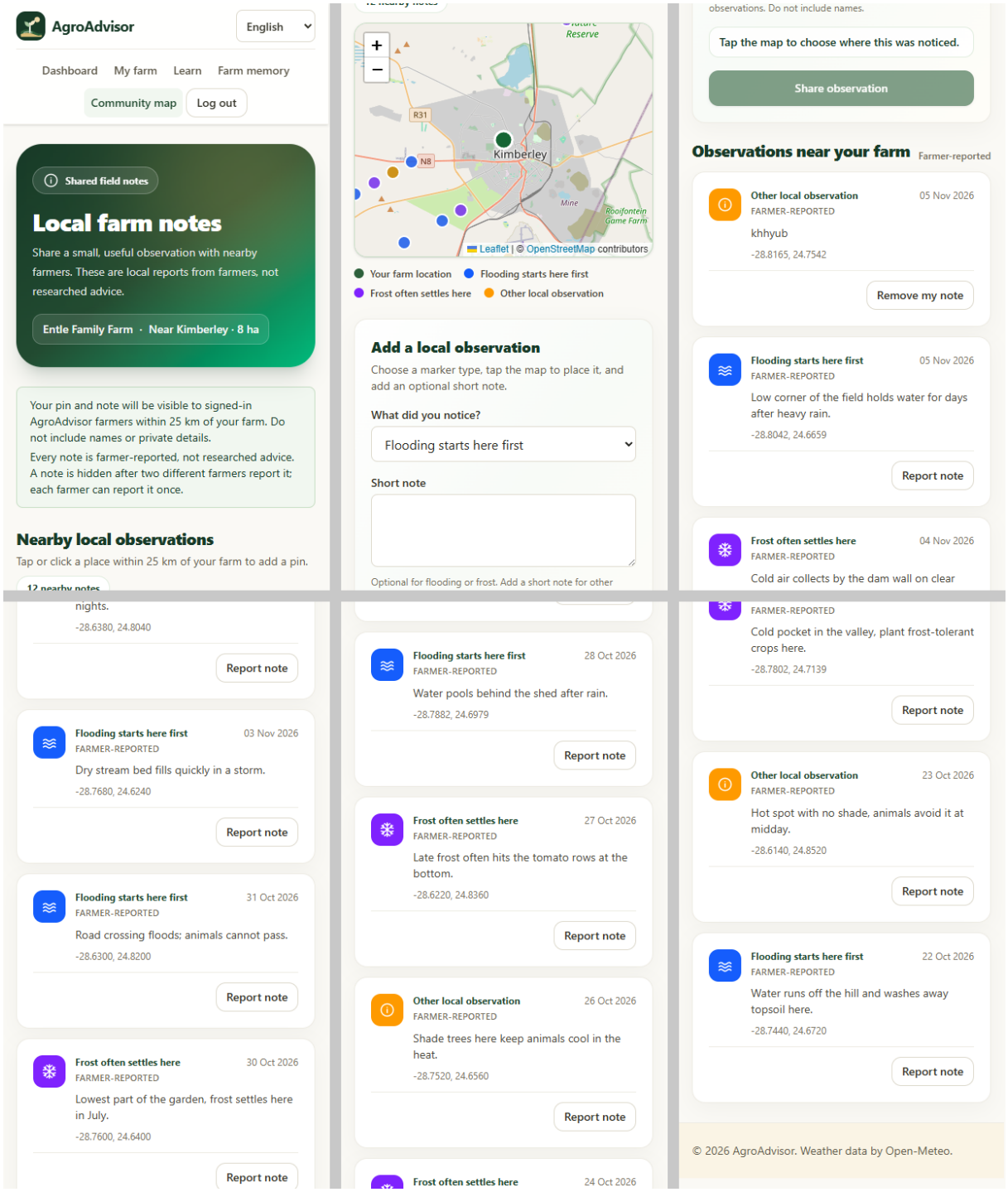


Figure: Community map: local field notes

- Purpose: farmers near each other share short local observations (flooding, frost or other).
- Shows: the farmer's own farm and nearby notes within 25 km on a map with a colour legend, a form to add an observation, and a list of nearby notes with date and place.
- Actions: tap the map to place a pin, choose the type, add an optional short note, Share observation; Report a note (another farmer's) or Remove my note. A note is hidden after two different farmers report it.

2. Database Design

AgroAdvisor stores its data in Cloud Firestore, a document database. Data is held in collections of documents instead of tables and rows, so the terms are used like this in this document:

- Entity = a collection (or sub-collection) of documents.
- Primary key (PK) = the document ID, which is unique inside its collection.
- Foreign key (FK) = a field that holds the ID of a document in another collection. Firestore does not enforce foreign keys, so the Security Rules and the application code keep them correct.
- Accounts (email, password, Google sign-in) are held by Firebase Authentication. The user ID (uid) it creates is the key that links all the data below to one farmer.

2.1 Entity relationship diagram

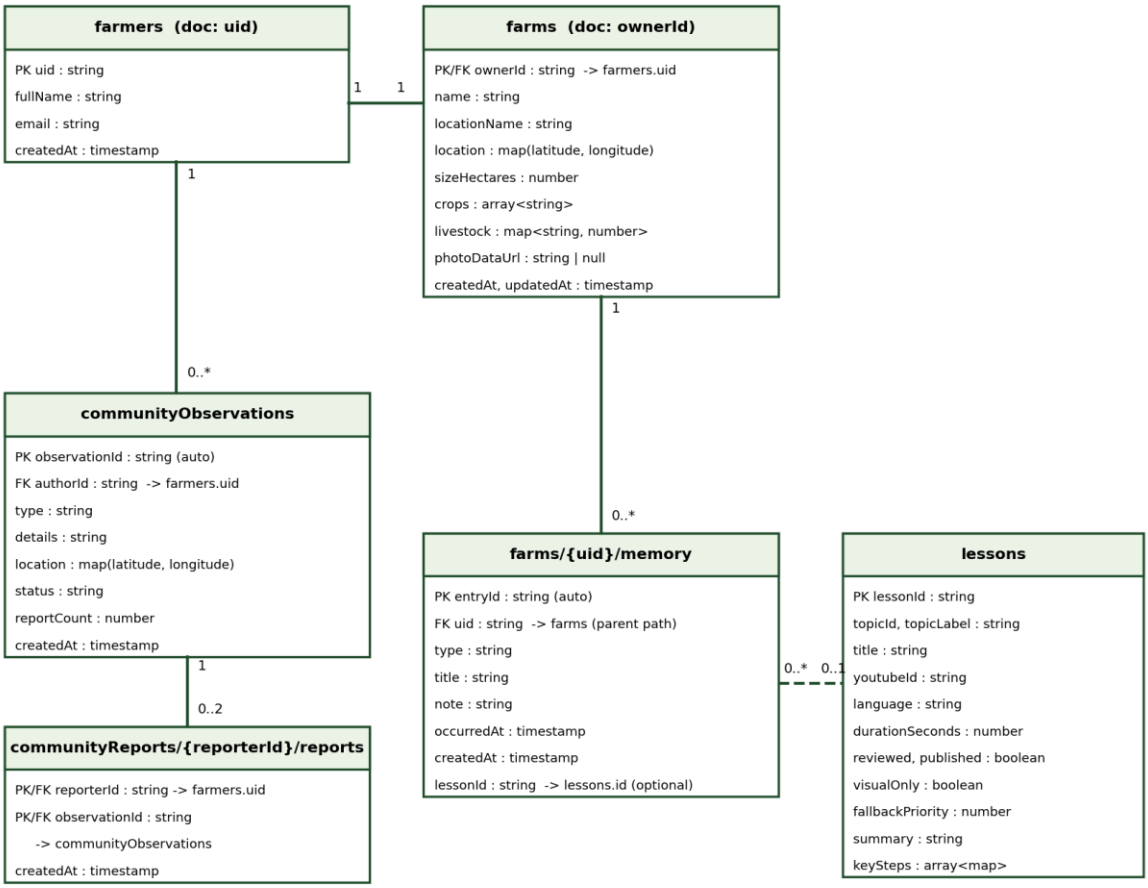


Figure: Entity relationship diagram of the Firestore collections

Relationship	Cardinality	How it is stored
farmers - farms	1 to 1	The farm document ID equals the farmer's uid, so a farmer has at most one farm.
farms - memory	1 to many	Memory entries are a sub-collection inside the farm document.
memory - lessons	many to 0..1	A memory entry of type 'lesson' keeps the lessonId of the lesson that was completed.
farmers - communityObservations	1 to many	authorId on the note holds the author's uid.

Relationship	Cardinality	How it is stored
communityObservations - communityReports	1 to 0..2	A report document is stored under the reporter's uid with the note's ID as its document ID. A note is hidden at two reports.

2.2 Entities, attributes and data types

2.2.1 farmers

Path: farmers/{uid}. One private profile per farmer. Only that farmer can read or write it.

Attribute	Data type	Key	Description and rules
uid	string	PK	Same as the Firebase Authentication user ID and the document ID.
fullName	string		Required, not empty.
email	string		Must equal the email of the signed-in account.
createdAt	timestamp		Set by the server when the profile is created and cannot change afterwards.

2.2.2 farms

Path: farms/{ownerId}. One farm per farmer. Only the owner can read or write it.

Attribute	Data type	Key	Description and rules
ownerId	string	PK, FK	References farmers.uid. Also the document ID.
name	string		1 to 80 characters.
locationName	string		Nearest town or village, 1 to 100 characters.
location	map { latitude: number, longitude: number }		Latitude -90 to 90, longitude -180 to 180.
sizeHectares	number		Greater than 0 and at most 10 000. The form can show acres.
crops	array of string		Crop and product names, at most 50.
livestock	map of string to number		Animal name and the number kept, at most 50 animal types.
photoDataUrl	string or null		Optional JPEG picture as a data URL, at most 100 000 characters.
createdAt, updatedAt	timestamp		Server times. updatedAt must be the time of the write.

2.2.3 memory (sub-collection of farms)

Path: farms/{uid}/memory/{entryId}. Private record of farm events. Entries cannot be edited, only created or deleted.

Attribute	Data type	Key	Description and rules
entryId	string	PK	Auto-generated document ID.
uid	string	FK	Taken from the parent path: the farm and farmer that own the entry.

Attribute	Data type	Key	Description and rules
type	string		One of weather, planting, success, lesson.
title	string		1 to 160 characters.
note	string		0 to 500 characters.
occurredAt	timestamp		When it happened. Cannot be in the future.
createdAt	timestamp		Server time of saving.
lessonId	string	FK	Only for type lesson: 1 to 200 characters, references lessons. Not allowed for other types.

2.2.4 lessons

Path: lessons/{lessonId}. Curated video lessons. Signed-in farmers can read only lessons with reviewed and published both true. Farmers cannot write lessons: the curator adds them in the Firebase console.

Attribute	Data type	Key	Description and rules
lessonId	string	PK	Document ID, for example garden-planning-basics.
topicId, topicLabel	string		Topic key and the topic title shown on the Learn page.
title	string		Lesson title.
youtubeId	string		Identifier of the YouTube video.
language	string		Language of the lesson.
durationSeconds	number		Length of the video.
sourceChannel	string		Name of the channel that made the video.
reviewed, published	boolean		Both must be true before farmers can see the lesson.
visualOnly	boolean		True when the video can be followed without speech.
fallbackPriority	number		Order used to choose a lesson when several match.
summary	string		One-line summary.
keySteps	array of map { icon: string, text: string }		Short steps with an icon name.

2.2.5 communityObservations

Path: communityObservations/{observationId}. Short local notes shared between farmers. Only visible notes can be listed.

Attribute	Data type	Key	Description and rules
observationId	string	PK	Auto-generated document ID.
authorId	string	FK	References farmers.uid. Must equal the signed-in user when the note is created. Only the author can delete the note.
type	string		One of flooding, frost, other.
details	string		At most 240 characters. Required when type is other.
location	map { latitude: number, longitude:		Must be within 25 km of the author's farm.

Attribute	Data type	Key	Description and rules
	number }		
status	string		visible when created; hidden after two reports.
reportCount	number (integer)		0 when created. Can only increase by 1 per report, to a maximum of 2.
createdAt	timestamp		Server time of creation.

2.2.6 communityReports (sub-collection)

Path: communityReports/{reporterId}/reports/{observationId}. Proof that one farmer reported one note, so nobody can report the same note twice. Only the reporter can read it, and it cannot be changed or deleted.

Attribute	Data type	Key	Description and rules
reporterId	string	PK, FK	Part of the path. References farmers.uid.
observationId	string	PK, FK	Document ID and field. References communityObservations. Together with reporterId it is unique.
createdAt	timestamp		Server time of the report.

2.3 Access rules summary

Entity	Read	Create	Update	Delete
farmers	Owner	Owner	Owner	Owner
farms	Owner	Owner	Owner	Owner
farms/memory	Owner	Owner	Nobody	Owner
lessons	Signed-in farmer (reviewed and published only)	Nobody (curator in console)	Nobody	Nobody
communityObservations	Signed-in farmer (visible notes)	Signed-in farmer with a farm, within 25 km	Only to add one report	Author
communityReports	Reporter	Reporter, once per note	Nobody	Nobody